

SmartEPT Admin Console — Installation Guide

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This guide takes a SmartEPT on-premise server from a bare machine to a licensed, running console with employees loaded and agents reporting. It covers Windows (both the quick Laragon route and a production IIS deployment), Linux and macOS.

Everything here refers to the package `SmartEPT-Admin-Server-Setup-1.1.zip`.

1. What you are installing

SmartEPT has three parts. This guide installs the first.

Part	What it is	Where it runs
Admin Console (this package)	The server: database, reports, policies, the <code>/admin</code> web console	One server inside the company
Desktop Agent	The tracking client that reports activity, breaks and attendance	Every employee PC
SmartEPT Central	Ametecs' licensing and client portal	Hosted by Ametecs — nothing to install

The agents talk to the Admin Console over your own network. Once licensed, an on-premise install **never needs the internet** — activation is offline and no data leaves the company.

2. Before you begin

2.1 Server requirements

	Minimum	Recommended
CPU	2 cores	4 cores
RAM	4 GB	8 GB
Disk	40 GB	100 GB+ (screenshots and webcam photos are the growth driver)
OS	Windows 10/11 Pro, Windows Server 2019+, Ubuntu 20.04+, RHEL 8+, macOS 12+	—
PHP	8.2 or newer	8.3
Database	MySQL 8 or MariaDB 10.5+	MySQL 8

Use PHP 8.3. PHP 8.5 is not recommended: several dependencies have not been validated against it, and SourceGuardian loaders lag well behind. Keep client servers on 8.3.

2.2 Required PHP extensions

`pdo_mysql`, `openssl`, `mbstring`, `curl`, `zip`, `gd`, `fileinfo`

The Linux and macOS installers check these for you and print the exact `apt` / `dnf` / `brew` command for anything missing. On Windows, enable them in `php.ini`.

2.3 Network

Direction	Port	Why
Employee PCs → server	TCP 80 / 443 (or 8080)	Agents report activity and fetch policy
Admin browsers → server	TCP 80 / 443 (or 8080)	The <code>/admin</code> console
Server → internet	not required	Only if you choose online licence validation

Give the server a **fixed IP or DNS name** before you install — it is baked into the agent configuration, and changing it later means reconfiguring every PC.

2.4 Decide these before you start

- The **company name** exactly as it should appear in reports
- The **admin email** for the first login
- Whether the company uses **biometric attendance** or **agent-only** attendance
- Your **shift timings** and break allowance

| 3. Windows — quick install (Laragon)

Best for a pilot, a small office, or an evaluation.

1. Install **Laragon Full** (bundles Apache, PHP and MySQL) and open it.
2. Click **Start All**. Confirm the traffic lights go green.
3. Unzip `SmartEPT-Admin-Server-Setup-1.1.zip` into `C:\laragon\www\`, giving you `C:\laragon\www\SmartEPT-Admin-Server\`.
4. Open that folder and double-click `INSTALL.bat`.
5. When prompted, enter your **admin email**, **company name** and a **password** (leave the password blank to have one generated and printed).
6. Open `http://smartept.test/admin` and sign in.

That is the whole install. Section 6 explains what those seven steps did.

| 4. Windows Server — production install with IIS

Use this when the console must run on port 80/443, start with Windows, and have no console window.

A. Prepare the server (once)

1. **IIS with CGI** — Server Manager → Add Roles and Features → *Web Server (IIS)* → Application Development → tick **CGI**. (*Windows 10/11 Pro: Turn Windows features on/off → IIS → Application Development Features → CGI.*)
2. **URL Rewrite module** — install from `https://www.iis.net/downloads/microsoft/url-rewrite`. This is not optional: the package ships a `public\web.config` that depends on it.
3. **PHP 8.3 NTS x64** — unzip from `windows.php.net/download` to `C:\PHP`, copy `php.ini-production` to `php.ini`, set `extension_dir = "ext"`, enable the seven extensions from §2.2, and add `C:\PHP` to the system PATH. Install the matching *Visual C++ Redistributable* if PHP complains on startup.
4. **Turn WebDAV OFF**. Do not skip this one — it bites every single time. IIS enables WebDAV by default and it intercepts the `PUT` and `DELETE` verbs before PHP sees them, so the console loads and creates records perfectly but **every Edit → Save and every Delete returns HTTP 405**. The shipped `web.config` removes it per-site, which is enough on most servers. If 405s persist, remove the feature: Server Manager → Remove Roles and Features → Web Server (IIS) → Common HTTP Features → untick **WebDAV Publishing** → then `iisreset`.
5. **MySQL 8 / MariaDB** — install it and note the credentials.

B. Install the application

6. Unzip the package to `C:\smartept`. Keep it **out of** `C:\inetpub\wwwroot` — only the `public` subfolder is ever exposed to the web.
7. Open an **administrator** Command Prompt in `C:\smartept` and run `INSTALL.bat`. If MySQL has a root password, edit `DB_USERNAME` / `DB_PASSWORD` in `.env` first and re-run.

C. Create the IIS site

8. IIS Manager → **Handler Mappings** (server level) → *Add Module Mapping*: Request path `*.php` · Module `FastCgiModule` · Executable `C:\PHP\php-cgi.exe` · Name `PHP-FastCGI` → OK → "create FastCGI application" → **Yes**.
9. **Sites** → **Add Website**: name `SmartEPT`, physical path `C:\smartept\public` — the `public` folder, never the app root. Port 80, plus an HTTPS binding with the company certificate if you have one. Application pool: **No Managed Code**.
10. **Permissions** — grant `IIS_IUSRS` Modify on the two writable folders (below).
11. In `.env` set `APP_URL=http://<server-name-or-ip>`, then run `php artisan config:clear` and `php artisan storage:link`.

Step 10 in full:

```
cd C:\smartept
icacls storage /grant "IIS_IUSRS:(OI)(CI)M" /T
icacls bootstrap\cache /grant "IIS_IUSRS:(OI)(CI)M" /T
```

Security check before go-live: browse to `http://<server>/.env`. If it downloads, the site is pointed at the app root instead of `public\`. Fix it immediately.

5. Linux and macOS

Linux (Ubuntu / Debian / RHEL family)

```
unzip SmartEPT-Admin-Server-Setup-1.1.zip
cd SmartEPT-Admin-Server
sudo bash install-linux.sh
```

The installer checks PHP and its extensions, prepares `.env`, creates the database, migrates, seeds roles, asks for your company and admin login, and — when run as root on a systemd system — installs and starts a `smartept-admin` service that survives reboot.

Console: `http://<server-ip>:8080/admin`

Useful commands afterwards:

```
sudo systemctl status smartept-admin
sudo systemctl restart smartept-admin
sudo journalctl -u smartept-admin -f
```

For a production Linux deployment, put nginx or Apache in front with the document root at `<app>/public` and a TLS certificate, rather than exposing port 8080 directly.

macOS

```
unzip SmartEPT-Admin-Server-Setup-1.1.zip
cd SmartEPT-Admin-Server
bash install-macos.sh
```

Prerequisites: `brew install php mysql && brew services start mysql`. The installer adds a launchd agent so the console starts at login. Console: `http://<mac-ip>:8080/admin`.

6. What the installer actually does

All three installers run the same seven steps through the same shared helper. Knowing them makes any failure easy to place.

Step	Action	Why it matters
1	Prepare <code>.env</code> , create the MySQL database	The stock config points at SQLite with no database file. This switches it to MySQL and creates the schema.
2	<code>composer install</code> — only if <code>vendor/</code> is missing	The package already ships <code>vendor/</code> , so this is normally skipped and no internet is needed.
3	<code>php artisan key:generate</code>	Without an app key every page is HTTP 500. The single most common cause of a failed setup.
4	<code>php artisan migrate</code>	Creates all 88 database tables.
5	Seed roles and permissions	Roles must exist before any admin can. Client servers get roles only — never demo data.
6	<code>storage:link</code> and clear caches	Links stored evidence and starts from a clean cache.
7	Create the first admin login	Without it there are no credentials to sign in with.

Step 1 also sets the on-premise licence posture: `SMARTEPT_ONPREM=true` and `SMARTEPT_LICENCE_ENFORCE=true`, which starts the 7-day evaluation and enables the `/activate` page.

7. The background scheduler — required

Several core features are **scheduled jobs, not live code**. Without the scheduler running, nightly attendance never completes, productivity summaries are never built, biometric punches never sync, meetings never auto-close, and the new post-shift auto sign-out never fires.

Windows — one Task Scheduler job, every minute, as SYSTEM:

```
schtasks /Create /TN "SmartEPT Scheduler" /SC MINUTE /MO 1 /RU SYSTEM ^
/TR "C:\PHP\php.exe C:\smartept\artisan schedule:run"
```

Linux / macOS — one crontab line:

```
* * * * * cd /path/to/SmartEPT-Admin-Server && php artisan schedule:run >> /dev/null 2>&1
```

Verify it: in the console, **Help → Troubleshooting**. The scheduler heartbeat turns red within minutes if `schedule:run` is not being called. Check it on every install.

What runs, and when

Job	Schedule	Purpose
<code>smartept:auto-logout</code>	every 5 min	New: signs out agents that never signed out, at shift end + N minutes
<code>smartept:biometric-sync</code>	every 5 min	Imports punches from cloud biometric devices
<code>smartept:close-meetings</code>	every minute	Ends meeting sessions at their scheduled end
<code>smartept:mark-attendance</code>	00:15	Completes yesterday's attendance sheet
<code>smartept:daily-summary</code>	00:30	Builds productivity summaries
<code>smartept:validate-license</code>	01:00	Daily licence check
<code>smartept:backup-database</code>	01:30	Nightly gzipped backup (keeps 14)
<code>smartept:purge-expired</code>	02:00	Applies your retention policy

8. Licensing and activation

1. Installation starts a **7-day evaluation** automatically — full features, no key needed.
2. To license, open `http://<server>/activate`. No sign-in is required.
3. Copy the **machine fingerprint** shown there and send it to Ametecs.
4. Ametecs issues a `.lic` file for that fingerprint. Upload it on the same page. Activation is **instant and fully offline** — the server never contacts the internet.
5. The licence is **node-locked**: it works only on the machine whose fingerprint it carries. If the server is replaced, Ametecs shifts the licence and issues a new `.lic`; the old machine stops validating and the history keeps both.
6. If the evaluation ends with no licence, monitoring stops but the console and `/activate` stay reachable, so you can license at any time.

Take the fingerprint **from the machine that will run in production**. A fingerprint from a test VM produces a licence that will not work on the real server.

9. First sign-in and initial setup

Work through this order — later steps depend on earlier ones.

9.1 Company

Organisation → Company. Set the company name, **timezone** (this drives every report boundary — get it right first), and the attendance source:

- **With biometric device** — the Gate-to-PC rule applies: an employee must punch at the door before their PC starts tracking.
- **Without biometric device** — agent login alone establishes attendance.

9.2 Organisation structure

Create in this order: **Branches → Departments → Teams → Designations → Shifts**. Bulk employee import creates any of these by name automatically, so if you are importing a prepared sheet you can skip ahead.

Shifts carry the fields the productivity report depends on:

Field	Notes
Start / End	Drives late arrival and early logout
Grace (min)	Late-arrival tolerance
Break allowed (min)	Pro-rated to actual presence in the report
Auto sign-out after shift end (min)	New in 1.1 — see §9.3

9.3 Post-shift auto sign-out — new in 1.1

If an employee never signs out, their session stays open and the productivity report has no sound end time to work from. Version 1.1 lets the server close the session itself.

Set it in **either** place:

- **Organisation → Shifts → Auto sign-out after shift end (min)** — per shift
- **Policies → Attendance → Auto sign-out after working end time (min)** — the fallback

The **shift value wins** when both are set. Blank in both means the feature is off, which is the default, so nothing changes until you configure it.

A sensible starting point is **60 minutes**. The sign-out is stamped at *shift end* + *N*, not at the moment the job runs, so a PC left on overnight is never credited the extra hours. A real punch-out or a manual HR correction always takes precedence.

9.4 Attendance policy

Policies → Attendance: late grace, early-logout grace, minimum working hours, attendance sources, and the auto sign-out fallback above. Assign the policy to the company (or to a branch/department/team for a narrower rule).

9.5 Monitoring, screenshot and webcam policies

Policies → Monitoring / Screenshot / Webcam. Assign each to the company, or lower down the tree for exceptions. Most specific wins: **Device > Employee > Team > Department > Branch > Company.**

Webcam presence and webcam photos are two separate settings:

Setting	What it does	Where it shows
<code>presence_enabled</code>	Detects presence locally and reports numbers only — never an image	Webcam tab → <i>Webcam presence detected</i>
<code>photo_enabled</code>	Captures and uploads actual photos	Webcam tab → <i>Webcam presence photos</i>

The shipped default is **presence on, photos off**. If you expect photos, tick `photo_enabled` explicitly.

Important: tracking mode **PRESENCE_ONLY** switches webcam presence **off**, despite the name. It means "presence at work" (attendance only), not face presence. For webcam presence the employee's tracking mode must be **FULL**.

9.6 Employees

Employees → Download CSV template, fill it, then **Bulk import (CSV)**.

Columns (in order):

```
employee_code, first_name, last_name, email, mobile, department, team, branch,
designation, shift, date_of_joining, biometric_id, employment_status, password
```

- Only `employee_code` and `first_name` are required.
- Department, team, branch, designation and shift are matched **by name** and created if new.
- `password` (**new in 1.1**) sets that employee's own login password, minimum 8 characters. Leave it blank to generate a temporary one that must be changed at first sign-in.
- Use **Preview (dry run)** first — it validates every row and changes nothing.
- **Export employees (CSV)** (new in 1.1) produces exactly these columns, so an export can be edited and re-imported. Passwords export blank, because they are stored hashed.

9.7 Users and roles

Users creates console logins (Company Admin, HR Admin, Branch Admin, Manager, Team Leader, Auditor, Compliance Officer). Employee self-service logins come from the import; console logins are separate and are created here.

| 10. Installing the desktop agent

1. From the client portal, download `SmartEPT-Agent-Setup-<version>.exe`.
2. Run it on each employee PC (per-machine install — administrator rights needed).
3. On first launch, enter the **server URL** — `http://<server>` or `http://<server>:8080`.
4. The employee signs in with the credentials from §9.6.
5. Confirm the PC appears under **Devices** in the console within a minute or two.

For a wide rollout, deploy the installer through your usual software-distribution tool and pre-seed the server URL with a `server.json` file beside the executable.

| 11. Upgrading an existing installation

1. **Back up first** — the database and the whole app folder.
2. Stop the site (IIS: stop the site; Linux: `sudo systemctl stop smartept-admin`).
3. Unzip the new package **over** the existing folder. Your `.env`, `license.lic` and `storage/` contents are not in the package and are left untouched.
4. Run the migrations (below). On Windows you can double-click `migrate.bat`, then `cache.bat`.
5. Start the site again and confirm **Help → Troubleshooting → System Health** is all green.

Step 4 in full:

```
php artisan migrate --force
php artisan optimize:clear
```

Version 1.1 adds one migration (post-shift auto sign-out) and is backward compatible — the new columns default to "off".

| 12. Backups

A nightly gzipped database backup runs automatically at 01:30 into `storage/app/backups`, keeping the newest 14. That protects the database only. **Also back up:**

- `.env` — your configuration and app key. **Without the app key, encrypted data is unrecoverable.**
- `license.lic` — your licence file
- `storage/app/` — stored screenshots, webcam photos and employee archives

Copy these off the server. A backup that lives only on the machine it protects is not a backup.

13. Troubleshooting

Start at **Help → Troubleshooting → System Health** inside the console. One click shows which component is red, with a fix link. **Help → Application log** has "Copy for developer" for support tickets.

Symptom	Cause	Fix
Every page is HTTP 500, "No application encryption key"	App key missing	Run <code>INSTALL.bat</code> , or <code>php artisan key:generate</code> then <code>cache.bat</code>
HTTP 405 on every Edit/Save and Delete, but pages load and Create works	WebDAV is intercepting PUT/DELETE	\$4 step 4, then <code>iisreset</code>
HTTP 500.19	URL Rewrite module missing	\$4 step 2
PHP files download instead of running	Handler mapping missing	\$4 step 8
"Could not connect / create the database"	MySQL down or wrong credentials	Start MySQL; check <code>DB_USERNAME</code> / <code>DB_PASSWORD</code> in <code>.env</code>
"pdo_mysql extension is not enabled"	Extension off	Enable <code>extension=pdo_mysql</code> in <code>php.ini</code> , restart
Login works but no data	<code>.env</code> still on SQLite	Set <code>DB_CONNECTION=mysql</code> , <code>DB_DATABASE=smartept</code> , re-run <code>INSTALL.bat</code>
"table doesn't exist" after an upgrade	Migrations not run	<code>migrate.bat</code> , or <code>php artisan migrate --force</code>
Cannot sign in on a brand-new install	Admin never created	<code>php artisan smartept:make-admin admin@company.com --company="Your Company"</code>
Changes never take effect	PHP OPcache frozen	Set <code>opcache.validate_timestamps=1</code> in <code>php.ini</code> , then a full Stop All → Start All
Attendance and reports stop updating overnight	Scheduler not running	\$7 — check the heartbeat in Help → Troubleshooting
Agent shows "punch in at the door"	Biometric gate is on	Organisation → Company → attendance source, or Gate Exclusion policy
Webcam presence shows nothing	Presence off, or tracking mode not FULL	\$9.5. On the PC, check <code>%APPDATA%\smartept-agent\presence.log</code>
<code>/storage/...</code> images 403 or 404	<code>storage:link</code> not run	<code>php artisan storage:link</code>

Reset an admin password

```
php artisan smartept:make-admin someone@company.com --company="Your Company"
```

Prints a fresh temporary password. Add `--password=YourPass` to set a specific one, or `--super` for a super-admin.

14. Support

Ametecs India Private Limited Email sales@ametecsindia.com · WhatsApp **90000 98877**

When reporting a problem, include: the SmartEPT version, the operating system and PHP version, what you were doing, and the output of **Help → Application log → Copy for developer**.

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